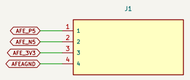
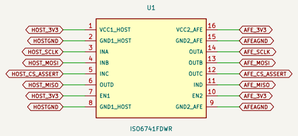
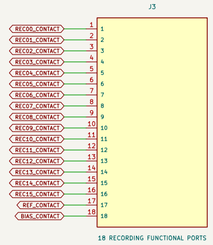
DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

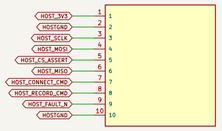
Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

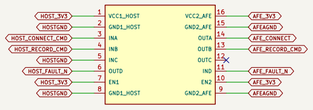




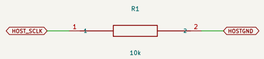










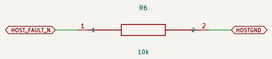




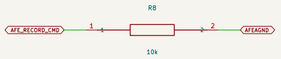








































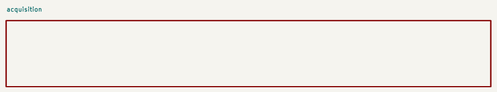


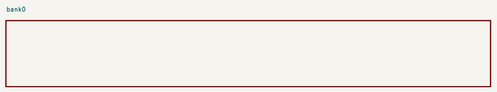


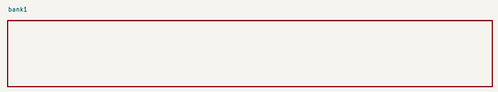


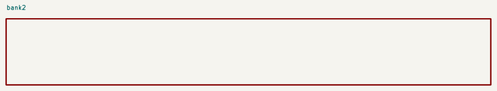


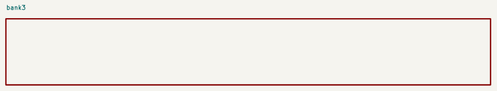
HOSTGND and AFEAGND are separate domains. Both receive valid regulated supplies externally; the digital barrier does not generate power.

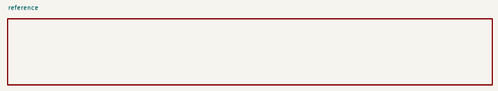












CS_ASSERT low -> island CS_N high. CONNECT low disconnects HV paths: ACQUIRE = CONNECT & RECORD, so 01 remains clamped.

Driver phase admission needs independently observed ISOLATED_READY; neither commands nor fault flags establish voltage or delivery evidence.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /

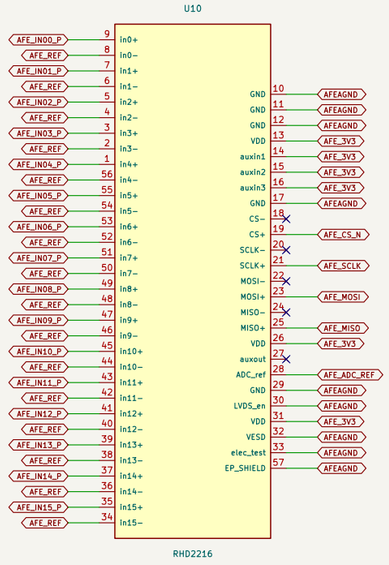
File: recording_front_end.kicad_sch

Title: ATO S4 / Isolated SPI, control and supplied power

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/7

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

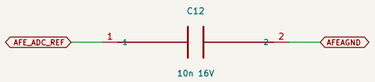
Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

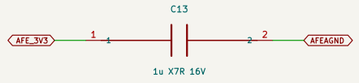


Every minus input shares AFE_REF. This is 16 differential streams from 18 functional recording ports, not 32 independent recording electrodes.









CMOS mode: LVDS_en=0; negative SPI pins NC; unused aux inputs at VDD; auxout NC; elec_test and VESD at AFEAGND.

Configure, calibrate and verify readback via SPI before valid data. 4MHz / 5us command slots are the revised isolated-link study baseline.

ADC response is pipelined; physical sample timestamps and invalid blanking/recovery intervals must be retained in software.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /acquisition/
File: acquisition.kicad_sch

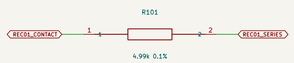
Title: ATO S4 / RHD2216 / sixteen sampled differential streams

Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/7

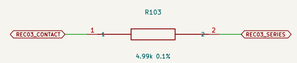
DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.







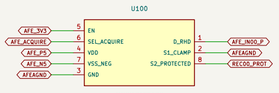


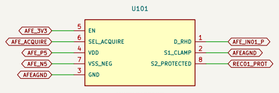


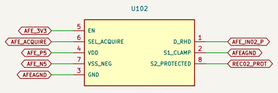


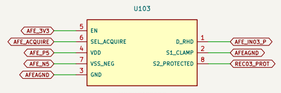




































Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2: S1 grounds RHD until ACQUIRE is admitted.







470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /bank0/
File: bank0.kicad_sch

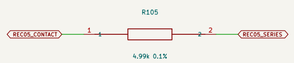
Title: ATO S4 / Inputs 00-03 / protected recording

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/7

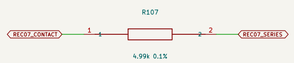
DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.







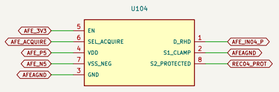


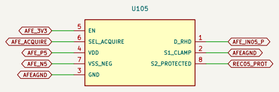


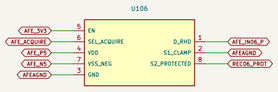


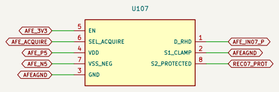


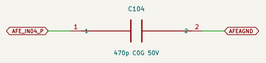


































Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2: S1 grounds RHD until ACQUIRE is admitted.







470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

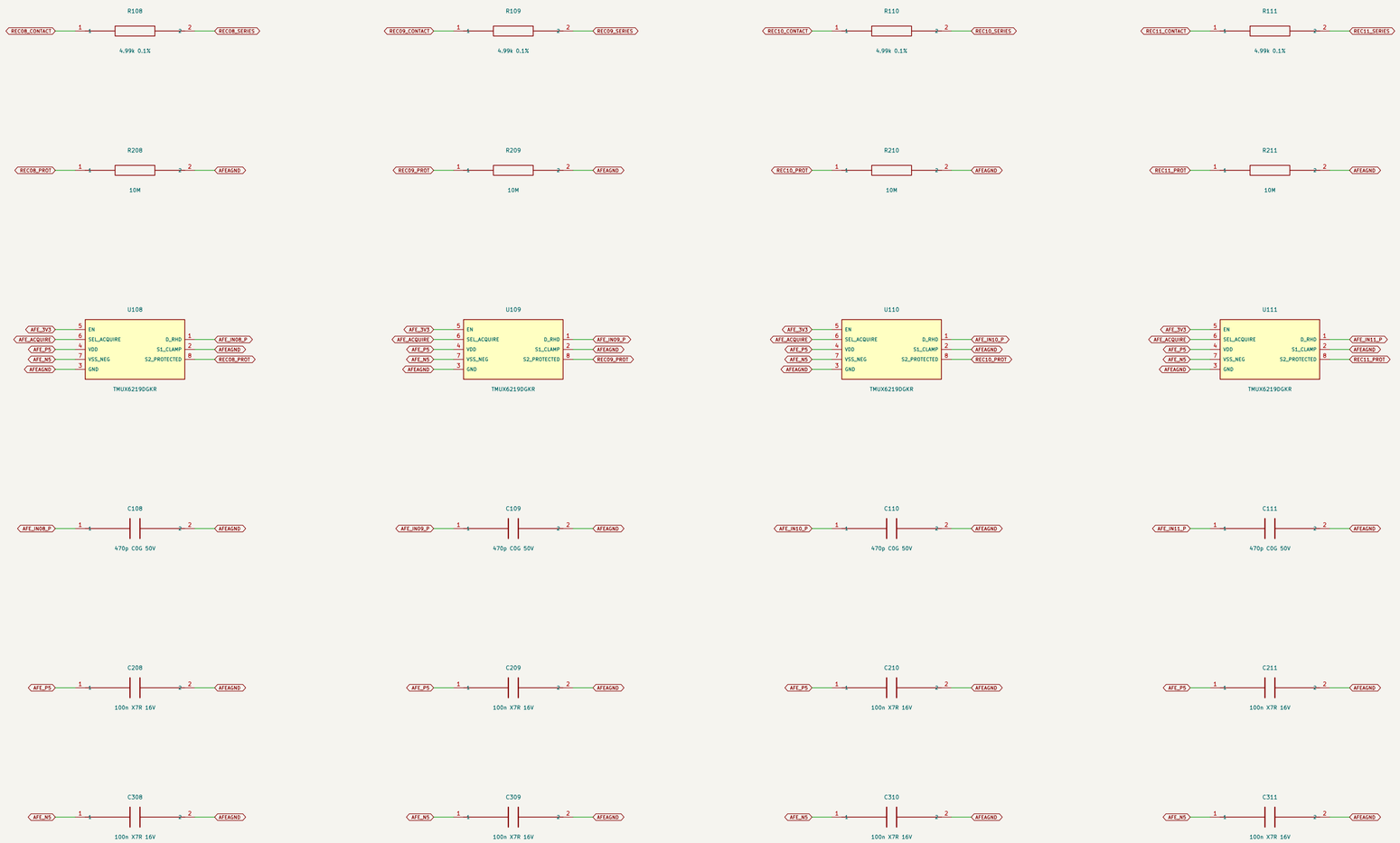
Sheet: /bank1/
File: bank1.kicad_sch

Title: ATO S4 / Inputs 04-07 / protected recording

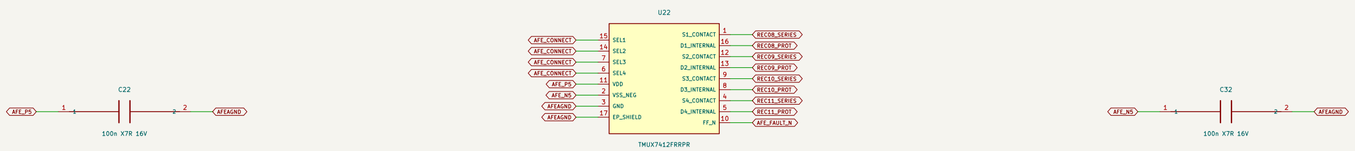
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/7

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.



Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2: S1 grounds RHD until ACQUIRE is admitted.



470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

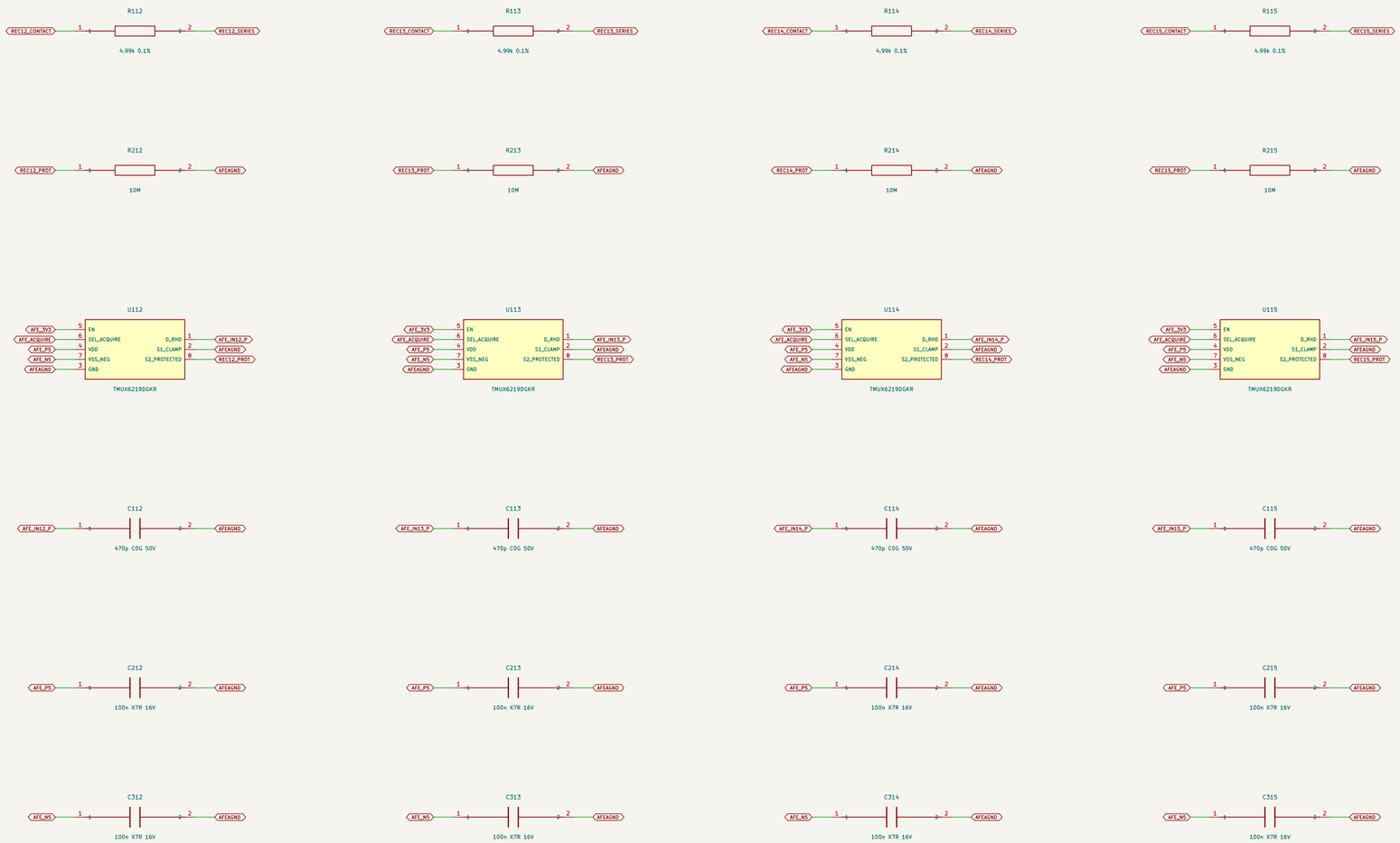
Sheet: /bank2/
File: bank2.kicad_sch

Title: ATO S4 / Inputs 08–11 / protected recording

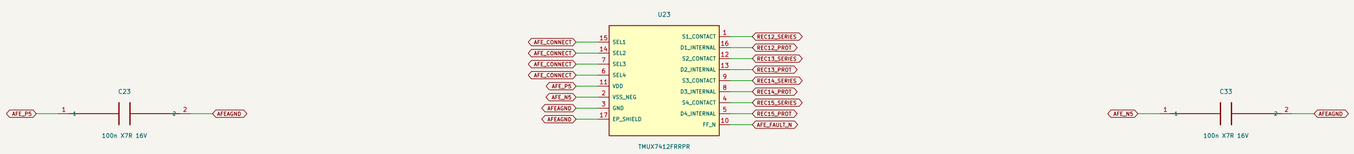
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/7

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.



Per column: contact -> 4.99k -> quad protected S/D -> protected node -> S2: S1 grounds RHD until ACQUIRE is admitted.



470pF is at the RHD pin; 10M bleeds the internal node. Switch capacitance/leakage/injection and shared reference mismatch remain modeled assumptions.

Fault threshold follows +/-5V rails and is far outside RHD +/-0.4V linear DC range. External low-voltage qualification is mandatory before unclamping.

Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /bank3/
File: bank3.kicad_sch

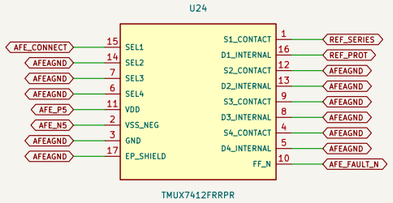
Title: ATO S4 / Inputs 12–15 / protected recording

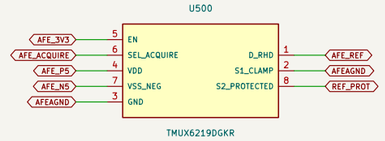
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/7

DIGITAL RESEARCH CIRCUIT / ARTIFICIAL SIGNALS / NO PCB OR IMPLANT QUALIFICATION

Spinal concept: 16 positive contacts + 1 shared reference + separate passive bias; amplifier linearity and biological recording remain conditional.

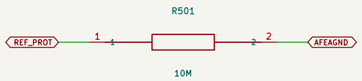
REF_CONTACT passes through the same disconnect/clamp topology; one AFE_REF node feeds all sixteen minus pins (192pF intrinsic typical).











BIAS_CONTACT is separate and never a stimulation return. Passive current arithmetic does not establish a medical current limit.













Native digital circuit slice; artificial signals only; external supervisors/power/PCB remain

Project ATO

Sheet: /reference/
File: reference.kicad_sch

Title: ATO S4 / Shared reference / passive body bias

Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 7/7